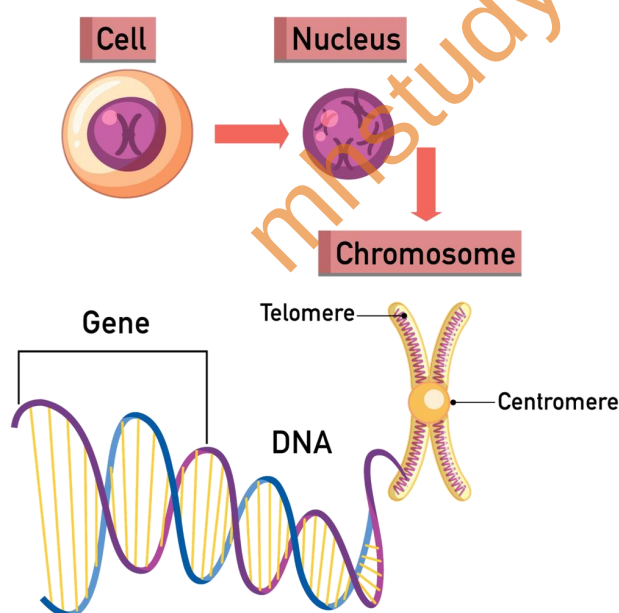




Chapter 2

LIFE

PROCESSES - 1



1. Fill in the blanks and EXPLAIN the statements.

- a. After complete oxidation of a glucose molecules, 38 number of ATP molecules are formed.
- b. At the end of glycolysis, Pyruvic acid, ATP, NADH₂ and water molecules are obtained.
- c. Genetic recombination occurs in pachytene phase of prophase-I of meiosis-I.
- d. All chromosomes are arranged parallel to equatorial plane of cell in metaphase phase of mitosis.
- e. For formation of plasma membrane, phospholipid molecules are necessary.
- f. Our muscle cells perform anaerobic type of respiration during exercise.

EXPLANATION

- a. We need the energy to work through the day, similarly each cell of our body needs the energy to survive. ATP is the energy currency in the cell. In cellular respiration glucose molecules break down (oxidation) in the presence of oxygen. On the complete breakdown, each glucose releases 38 ATP molecules.
- b. A molecule of glucose is oxidized step by step in the process of glycolysis where two molecules of each, i.e. pyruvic acid, ATP, NADH₂ and water are formed.

- c. During Meiosis-I, in pachytene stage, crossing over, i.e. exchange of genetic material takes place between non-sister chromatids of homologous chromosomes.
- d. Nuclear membrane completely disappears in metaphase. Chromosomes completely condense and become clearly visible along with sister chromatids. All chromosomes are parallelly arranged to the equatorial plane of the cell.
- e. Phospholipids which are essential for producing plasma membrane are formed from fatty acids. They form the basic structure of plasma membrane and are essential for its semi-permeability.
- f. While exercising, the oxygen supply may not be enough for the need of muscles. Hence, muscle cells perform anaerobic respiration. As a result, less amount of energy is produced in our body and lactic acid accumulates due to which we feel tired.

2. Write definitions.

- a. Nutrition - Nutrition is the process of taking in food and converting it into energy and other vital nutrients required for life
- b. Nutrients - Nutrients are components of food which provide energy and help in growth and maintenance of the body.

- a. **Proteins** - Proteins are the macromolecules formed by bonding together many amino acids.
- b. **Cellular respiration** - Cellular respiration is a process where foodstuffs are oxidized step by step in the cells. Cellular respiration occurs among the living organisms by two methods. Those two methods are aerobic respiration (oxygen is involved) and anaerobic respiration (oxygen is not involved).
- c. **Aerobic respiration** - Aerobic respiration is the process of obtaining energy from foodstuffs with the help of oxygen.
- d. **Glycolysis** - Glycolysis is the process in which a molecule of glucose is oxidized in a step-by-step process and two molecules of each, i.e. Pyruvic acid, ATP, NADH₂ and water are formed .

3. **Distinguish between**

a. **Glycolysis and TCA cycle.**

Glycolysis	TCA cycle
1. Glycolysis occurs in the cytoplasm.	1. TCA cycle occurs in the mitochondria.
2. Two molecules each of Pyruvic acid, ATP, NADH ₂ and water are formed.	2. Molecules of CO ₂ , H ₂ O, NADH ₂ and FADH ₂ are formed.

3. It is also called as EMP pathway.
4. No carbon dioxide is released.

3. It is also called as Kreb's cycle.
4. Carbon dioxide is released.

b. Mitosis and Meiosis

Mitosis	Meiosis
1. Mitosis takes places in the somatic cells. 2. Occurs in both sexually as well as asexually reproducing organisms. 3. The cell divides only once. 4. Chromosome number remains constant at the end of Mitosis.	1. Meiosis takes place in the germ cells. 2. Occurs only in sexually reproducing organisms. 3. There are two cell divisions, Meiosis I and Meiosis II. 4. Chromosome number is reduced from diploid to haploid, at the end of Meiosis.

c. Aerobic and Anaerobic respiration

Aerobic Respiration	Anaerobic Respiration
1. Process occurs in the presence of oxygen. 2. Glucose is completely oxidised.	1. Process occurs in the absence of oxygen. 2. Glucose is incompletely oxidised.

3. Energy is released in greater amounts in the form of ATP.
4. The end products are CO_2 and water.

3. Energy is released in lesser amounts in the form of ATP.
4. The end products are CO_2 and organic acids or alcohol.

4. Give scientific reasons.

a. Oxygen is necessary for complete oxidation of glucose.

- i. Glucose is completely oxidized in aerobic respiration to produce 38 molecules of ATP.
- ii. Aerobic respiration consists of three steps. These are Glycolysis, Krebs cycle and Electron Transport Chain reaction.
- iii. In absence of oxygen, i.e., in anaerobic respiration only glycolysis takes place. Glucose is incompletely oxidized and very less amount of energy (2 ATP) is produced.
- iv. Therefore, oxygen is necessary for complete oxidation of glucose.

b. Fibers are one of the important nutrients.

- i. We obtain fibers from leafy vegetables, fruits, cereals, etc.
- ii. We cannot digest the fibers.
- iii. However, they help in the digestion of other substances and egestion of undigested substances.
- iv. Hence, fibers are also an essential nutrient.

c. Cell division is one of the important properties of cells and organisms.

- i. There are two types of cell division as mitosis and meiosis. Mitosis occurs in somatic cells and stem cells of the body whereas meiosis occurs in germ cells.
- ii. Due to the property of cell division, a new organism is formed from existing one, a multicellular organism grows up and emaciated body can be restored.
- iii. It is required for wound healing and formation of blood cells.
- iv. Thus, cell division is one of the important properties of cells and living organisms.

d. Sometimes, higher plants and animals too perform anaerobic respiration.

- i. Some higher plants, animals and aerobic microorganisms also perform anaerobic respiration instead of aerobic respiration if there is depletion in oxygen level in the surrounding.
- ii. Glucose is incompletely oxidized and less amount of energy is obtained in anaerobic respiration.
- iii. Seeds perform anaerobic respiration if the soil is submerged under water during germination.

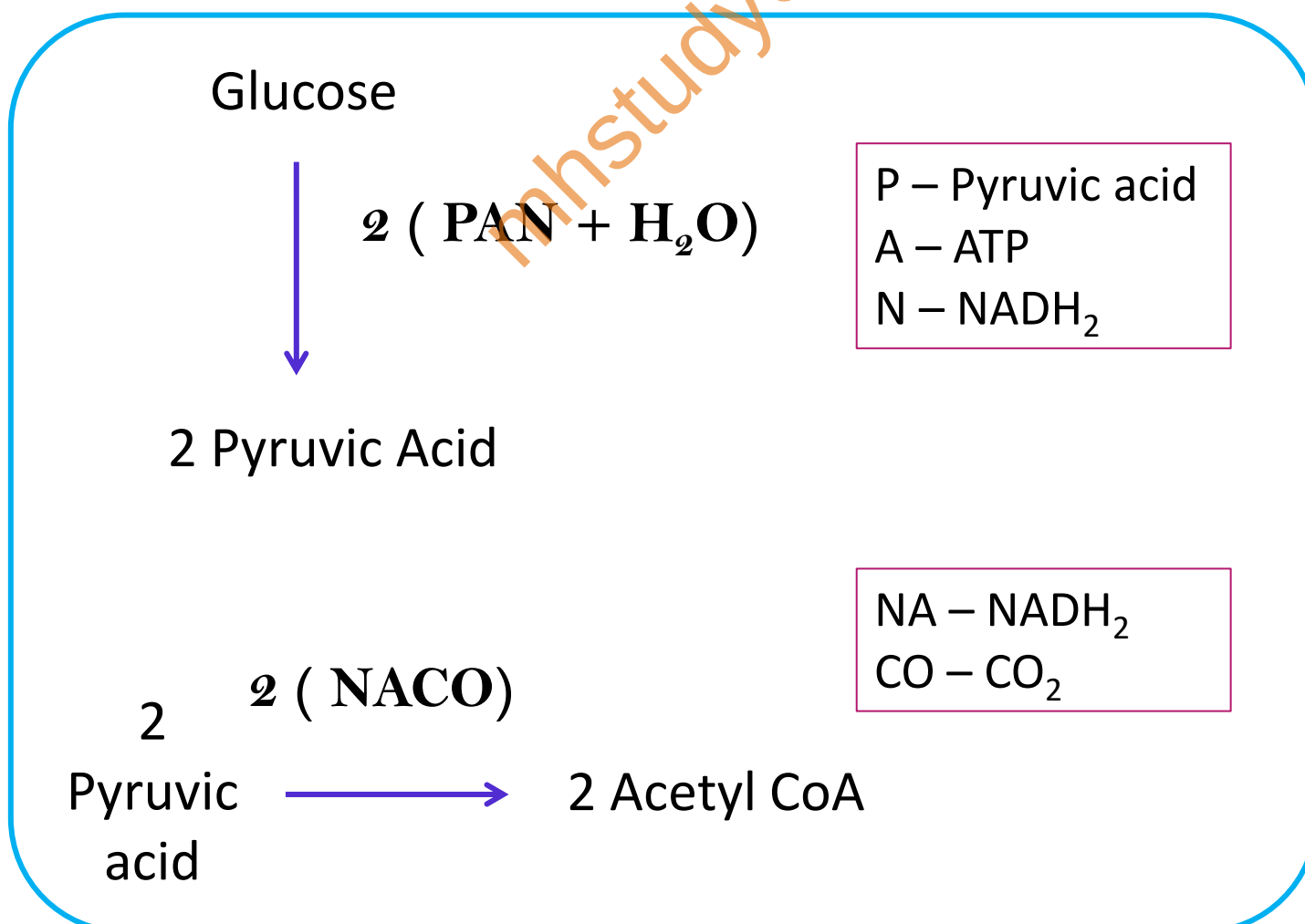
- iv. Similarly, our muscle cells also perform anaerobic respiration while exercising. Less amount of energy is produced in our body and lactic acid accumulates due to which we feel tired.

e. Kreb's cycle is also known as citric acid cycle.

- i. Acetyl-CoA (2 carbon compound) enters the mitochondria to combine with oxaloacetic acid (4 carbon compound) and form a six-carbon compound i.e., citric acid.
- ii. Since citric acid is the first stable compound formed, Krebs cycle is also called the citric acid cycle.

5. Answer in detail.

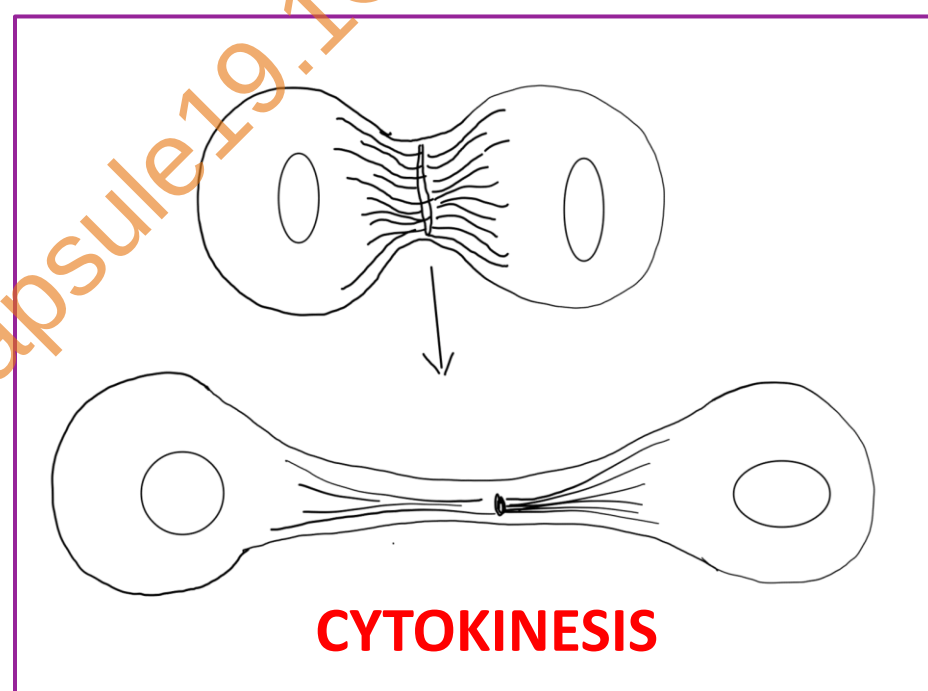
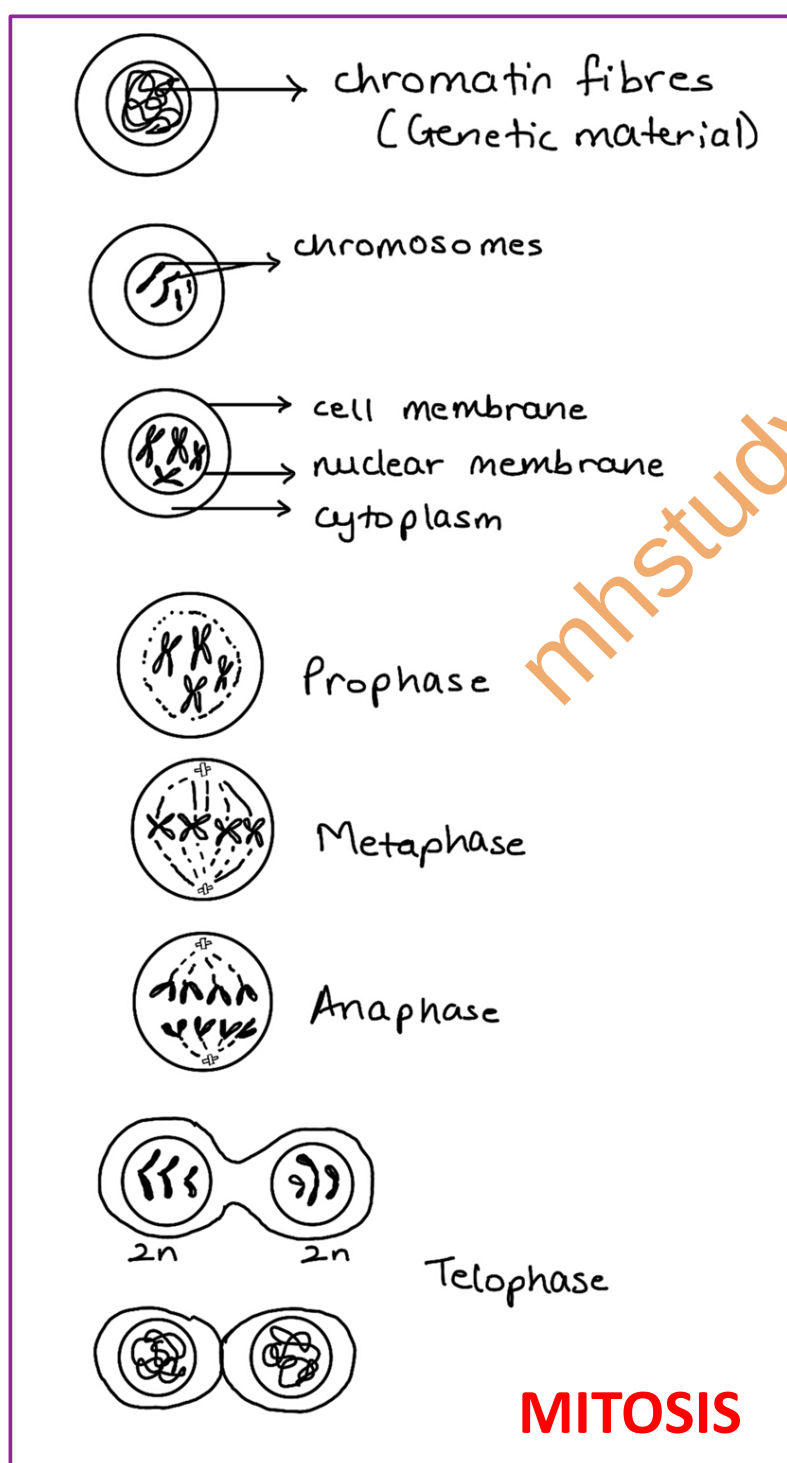
a. Explain the glycolysis in detail.



Just to
memorise.
Not to be
written in
exam.

- i. Process of glycolysis occurs in cytoplasm.
- ii. A molecule of glucose is oxidized step by step in this process and two molecules of each i.e., pyruvic acid, ATP, NADH_2 and water are formed.
- iii. Molecules of pyruvic acid formed in this process are converted into molecules of Acetyl-Coenzyme-A.
- iv. Two molecules of NADH_2 and two molecules of CO_2 are released during this process.

b. With the help of suitable diagrams, explain the mitosis in detail.



Both diagrams to be drawn in exam.

P - Prophase

M - Metaphase

A - Anaphase

T - Telophase

1. Somatic cells and stem cells divide by mitosis.
2. Mitosis is completed through two main steps,
 - a. Karyokinesis (nuclear division)
 - b. Cytokinesis (cytoplasmic division)
3. Karyokinesis is completed through 4 steps.

A. Prophase

- i. In prophase, condensation of basically thin thread-like chromosomes starts.
- ii. Due to this, they become short and thick and they start to appear along with their pairs of sister chromatids.
- iii. Centrioles duplicate and move to opposite poles of the cells.
- iv. Nuclear membrane and nucleolus start to disappear.

B. Metaphase

- i. Nuclear membrane completely disappears in metaphase.
- ii. Chromosomes complete their condensation and become clearly visible along with their sister chromatids.
- iii. All chromosomes are arranged parallel to equatorial plane (central plane) of the cell.
- iv. Special type of flexible protein fibers (spindle fibers) are formed between centromere of each chromosome and centrioles.

C. Anaphase

- i. In anaphase, centromeres split and thereby sister chromatids of each chromosome separate and they are pulled apart in opposite directions with the help of spindle fibers.
- ii. Separated sister chromatids are called as daughter chromosomes.
- iii. Chromosomes being pulled apart appear like bunch of bananas.
- iv. In this way, each set of chromosomes reach the two opposite poles of the cell.

D. Telophase

- i. The chromosomes which have reached at opposite poles of the cell now start to decondense and again become thread-like thin and invisible.
- ii. Nuclear membrane is formed around each set of chromosomes reached at opposite poles. Thus, two daughter nuclei are formed in a cell.
- iii. Nucleolus also appears in each daughter nucleus.
- iv. Spindle fibers completely disappear.

4. In this way, karyokinesis completes and cytokinesis begins.

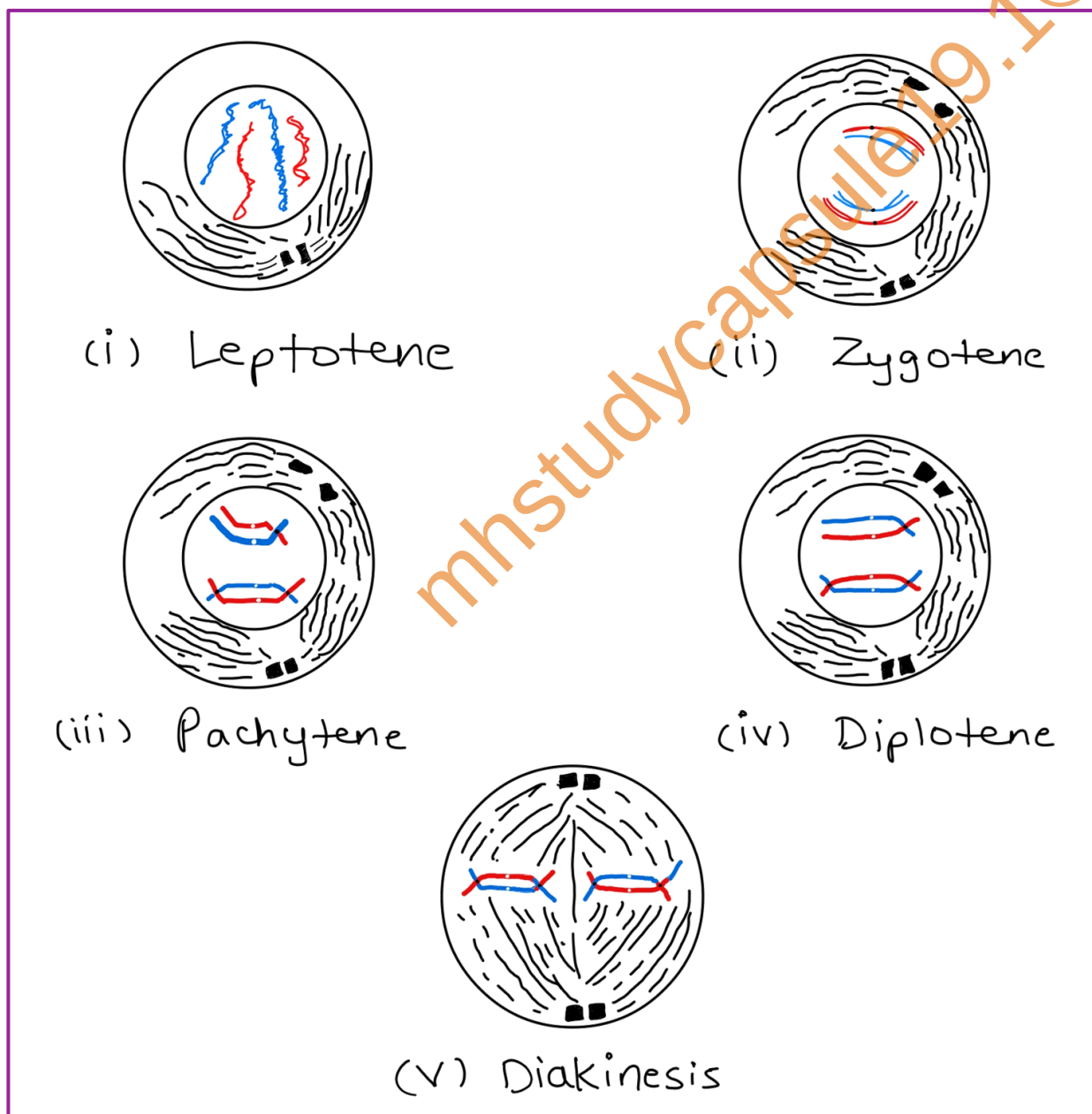
5. Cytokinesis :

- i. The cytoplasm divides by cytokinesis and two new cells are formed which are called as daughter cells.

- ii. In this process, a notch is formed at the equatorial plane of the cell which deepens gradually and thereby two new cells are formed.
- iii. However, in case of plant cells, instead of the notch, a cell plate is formed exactly along midline of the cell and thus cytokinesis is completed.

c. With the help of suitable diagrams, explain the five stages of prophase-I of meiosis.

NEVER ASKED IN EXAM. YOU CAN SKIP THE QUESTION.



Le

Zy

Pa

- i. Prophase I is the most important stage of meiosis and differs considerably from the prophase of mitosis.
- ii. It is divided into 5 stages leptotene, zygotene, pachytene, diplotene and diakinesis.
- iii. It involves essentially the crossing over and recombination of genetic material between non-sister chromatids.
- iv. Leptotene: In this stage already replicated chromatin network condenses and start converting into long thin thread like chromosomes.
- v. Zygotene: In this stage homologous chromosomes come closer to each other. This pairing is called 'Synapsis'.
- vi. Pachytene: Sister chromatids become visible in this stage. Crossing over occurs between non-sister chromatids of the homologous chromosomes. The crossing over is called as 'Chiasma or Chiasmata' and this leads to genetic recombination causing variations.
- vii. Diplotene: In this stage, recombined sister chromatids start repelling. Nucleolus and nuclear membrane start disappearing.
- viii. Diakinesis : Chromosomes are fully condensed. Nucleolus and nuclear membrane completely disappear, spindle fibres begin to appear.

d. How all the life processes contribute to the growth and development of the body?

Life processes contribute to growth and development of the body in the following way:

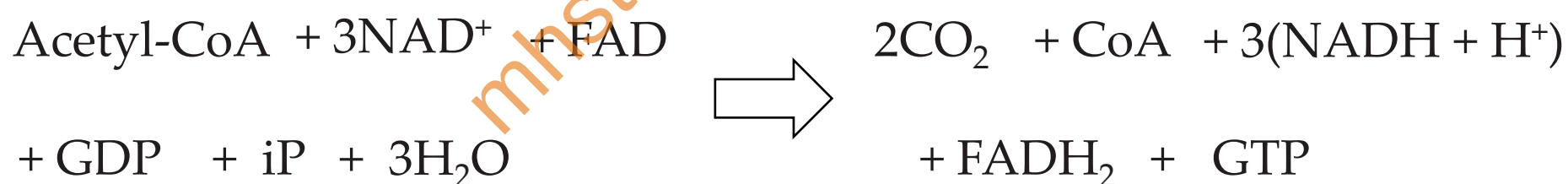
- i. Metabolism: Metabolism provides the energy and building blocks necessary for the growth and development of new tissues and organs. It involves both catabolic (energy-releasing) and anabolic (building) reactions.
- ii. Cell Division and Reproduction: Cell division allows the body to produce new cells, which then differentiate and specialize to form various tissues and organs. Reproduction ensures the continuation of the species and the passing on of genetic information.
- iii. Nutrition and Digestion: Proper nutrition supplies essential nutrients, vitamins, and minerals required for tissue growth, repair, and maintenance. Digestive processes break down food to extract these nutrients.
- iv. Hormones and Endocrine System: Hormones regulate growth and development processes. For example, growth hormone stimulates tissue growth, and sex hormones influence sexual maturation and reproductive development.

- v. Circulatory System: The circulatory system transports nutrients, oxygen, hormones, and other essential substances throughout the body, ensuring that all cells receive the necessary resources for growth.

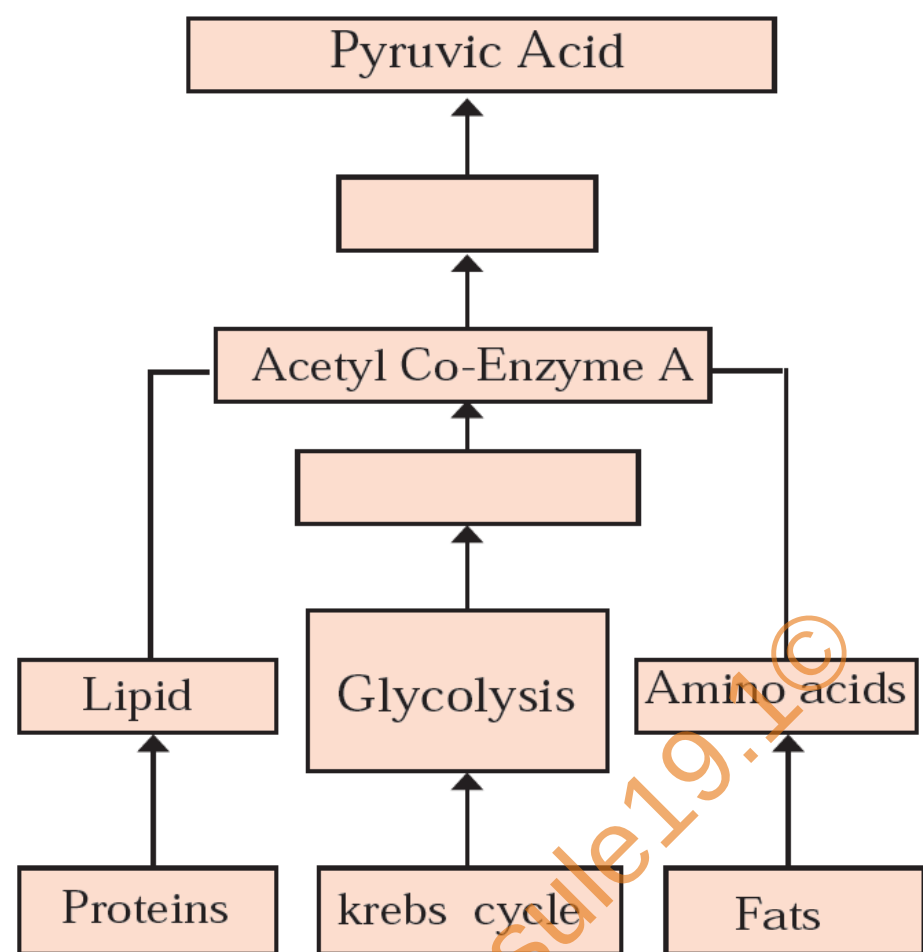
These interconnected life processes work together harmoniously to support the growth, maturation, and proper functioning of the human body.

e. Explain the Kreb's cycle with reaction.

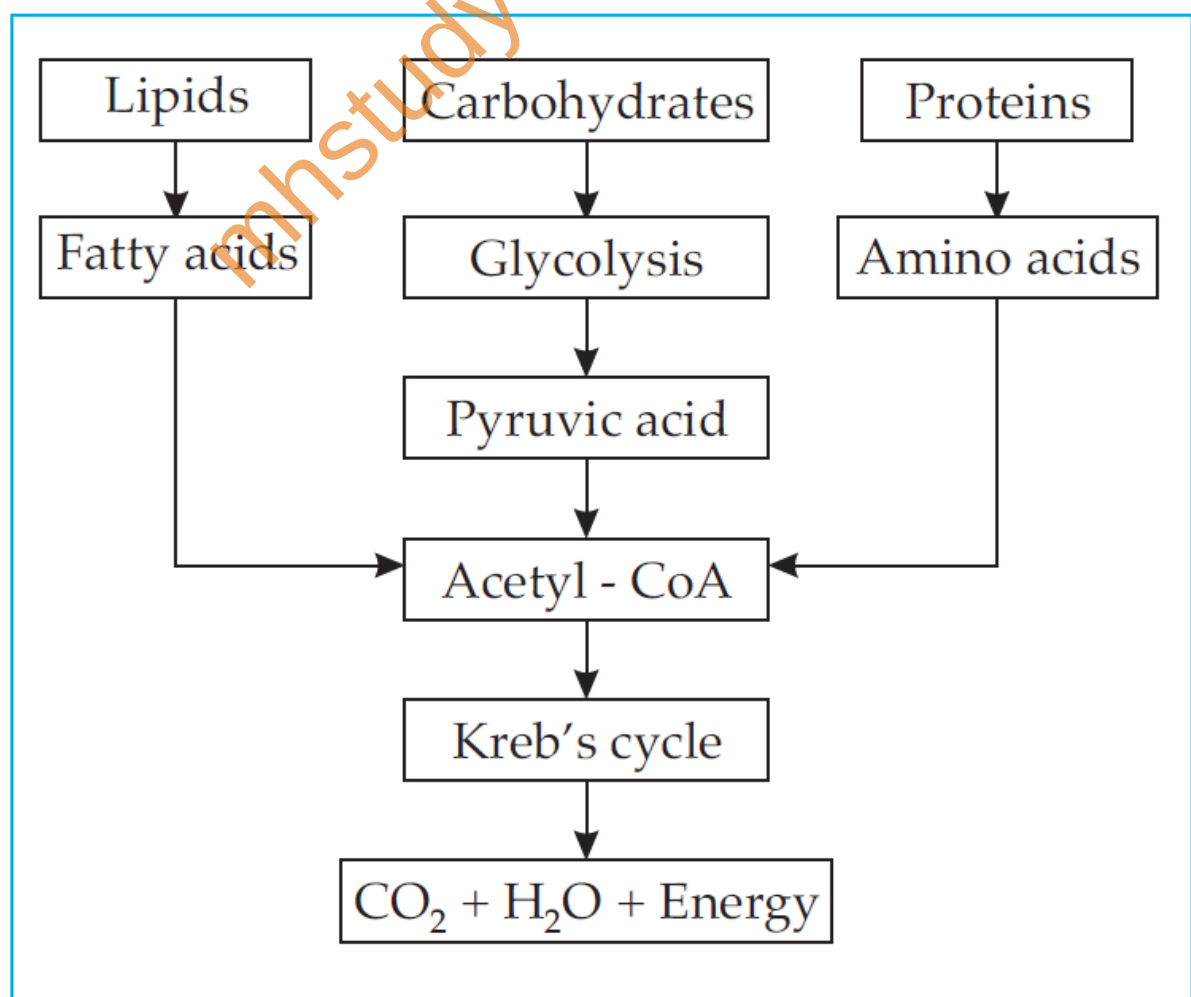
- i. Both molecules of acetyl-CoA enter the mitochondria.
- ii. Cyclic chain of reactions called as tricarboxylic acid cycle is operated on it in the mitochondria.
- iii. Acetyl part of acetyl-CoA is completely oxidized through this cyclical process and molecules CO_2 , H_2O , NADH_2 , FADH_2 are derived.
- iv. Net reaction of Kreb's cycle:



5. How is energy formed from oxidation of carbohydrates, fats and proteins? Correct the diagram given below.



Ans.



INTEXT QUESTIONS

1. How are the food stuffs and their nutrient contents useful for body?

- i. The food stuffs are digested and converted into soluble nutrients.
- ii. These nutrients are carried by blood to every cell of the body.
- iii. The oxygen inhaled at the time of respiration is also carried to every cell.
- iv. In the body cells, this oxygen carries out oxidation of nutrients and thus energy is produced.
- v. The energy helps the body to carry out all its functions.
- vi. The nutrients help in the growth and development of the body.

2. What is the importance of balanced diet for body?

- i. A balanced diet is a diet containing all nutrients (carbohydrates, fats, proteins, vitamins, etc.) in the right quantities.
- ii. A balanced diet helps us in various ways:
 - Provides us with important nutrients required for the growth of body.
 - Helps us to stay fit and free from diseases.
 - Helps in strengthening the immune system.
 - Helps our systems to function properly.
 - Helps maintain a healthy weight.
 - Provides the body with energy.

3. Which different functions are performed by muscles in body?

- i. Muscles bring about movement of arms and legs, running, speaking, walking etc.
- ii. Muscles also bring about movement of eyelids, passage of food through the alimentary canal, contraction and relaxation of blood vessels.
- iii. Cardiac muscles bring about contraction and relaxation of the heart.

4. What is the importance of digestive juices in digestive system?

- i. Digestive juices create conditions required for digestion of food.
- ii. Digestive juices contain enzymes which break down different components of food like carbohydrates, proteins and fats, so that the essential nutrients are absorbed and properly utilized by the body.

5. Which system is in action for removal of waste materials produced in human body?

Excretory system is in action for removal of waste materials produced in human body.

6. What is the role of circulatory system in energy production?

The circulatory system allows the transport of energy rich substances and oxygen in the blood to the body's cells so that they can produce energy for their functioning.

7. How are the various processes occurring in human body controlled? In how many ways ?

- i. Various processes occurring in the human body is controlled in two ways, and the two organ systems responsible for the control and coordination are the nervous system and the endocrine system.
- ii. The nervous system: This system involves all types of nerves and organs like the brain and spinal cord. The function of this organ system is to coordinate the function of all the internal and external organ through electrical impulses. The brain controls every physiological functions hence is called the command centre of the human body.
- iii. The endocrine system: This system consists of endocrine glands and hormones. It controls many physiological functions in our body like growth, development, reproduction, metabolism, etc., by secreting hormones to the bloodstream.

8. Why do players consume food stuffs during breaks of the game?

- i. Players are in need of continuous energy during games.
- ii. Carbohydrates (glucose) are a major and quick source of energy.
- iii. Hence, players are seen consuming food stuffs rich in carbohydrates during breaks of the game.

9. What is respiration? How does it occur?

- i. The process of release of energy by oxidation of food is called as respiration.
- ii. In living organisms, respiration occurs at two levels - body and cellular level.
- iii. In case of body level, oxygen and carbon dioxide are exchanged between the body and surrounding.
- iv. In case of cellular level, food is oxidized either with or without the help of oxygen to release energy.

10. How many atoms of C, H and O are respectively present in a molecule of glucose?

There are 6 atoms of C, 12 atoms of H and 6 atoms of O present in a molecule of glucose ($C_6H_{12}O_6$).

11. Which types of chemical bonds are present between all atoms in a molecule of glucose?

Covalent bonds are present between all atoms in a molecule of glucose.

12. In terms of Chemistry what happens actually when a molecule is oxidized?

In terms of chemistry when a molecule is oxidized it gains oxygen or loses electrons or hydrogen.

13. Observe and Label the diagram given beside.

Nasal cavity →

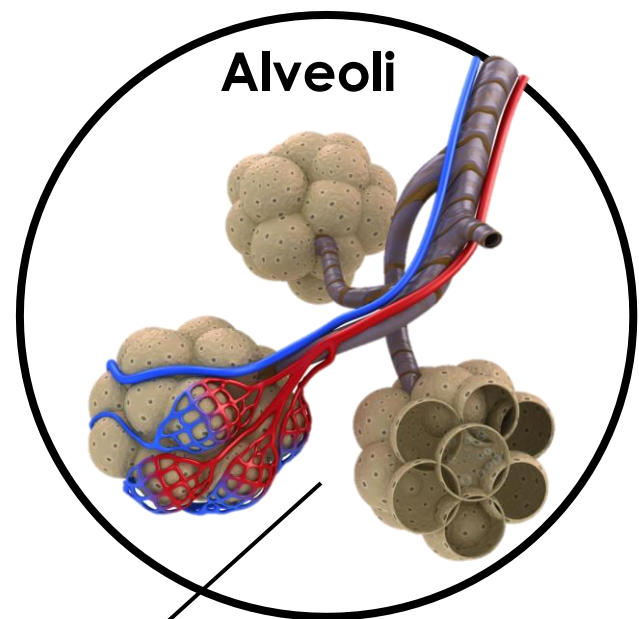
Oral cavity →

Larynx →

Trachea →

Right lung →

Diaphragm →



Left primary
bronchus



Always Remember.

NAD - Nicotinamide Adenine
dinucleotide

FAD - Flavin adenine dinucleotide

Both coenzymes are formed in the cells
and used in cellular respiration.

14. Which type of cellular respiration performs complete oxidation of glucose?

Aerobic respiration performs complete oxidation of glucose.

15. Which cell organelle is necessary for complete oxidation of glucose?

Mitochondria is necessary for complete oxidation of glucose

16. From where do we obtain the lipids?

We obtain lipids from milk, butter, cheese, oil, ghee, meat, nuts and oilseeds, etc.

17. Many times, you cannot eat hot food due to inflammation/ulceration in mouth.

- i. One of the reason for ulcers in mouth is deficiency of Vitamin B₁₂.
- ii. Inflammation or ulceration in the mouth can lead to heightened sensitivity and pain when exposed to hot temperatures.
- iii. Hot foods may aggravate the condition, causing further discomfort or irritation.
- iv. Therefore, many times, you cannot eat hot food due to inflammation/ulceration in mouth.

18. Some persons experience difficulty in night vision since their childhood or adolescence.

- i. The difficulty in night vision is called night blindness.

- ii. Night vision relies on the proper functioning of the retina, the light-sensitive tissue at the back of the eye.
- iii. Vitamin A is crucial for maintaining healthy retinal cells, and a deficiency in this vitamin can lead to impaired night vision.
- iv. As a result, some persons experience difficulty in night vision.

19. Many times, we experience dryness in mouth.

One of the most common causes of dry mouth is dehydration. Not consuming enough fluids can lead to a decrease in saliva production, resulting in a dry and uncomfortable sensation in the mouth.

20. Oral rehydration solution (Salt-sugar-water) is frequently given to persons experiencing loose motions.

- i. ORS is a simple and effective way to rehydrate the body when a person is suffering from diarrhea.
- ii. Diarrhea can lead to significant fluid loss, which can result in dehydration.
- iii. ORS contains a balanced combination of salt, sugar, and water, which helps the body to absorb the lost fluids and electrolytes more efficiently than water alone.

21. We sweat during summer and heavy exercise.

- i. Human beings are warm-blooded. We need to maintain a constant body temperature irrespective of the surrounding conditions.
- ii. In summer and during heavy exercise, lot of heat is generated which raises our body temperature.

- iii. Sweating leads to loss of excess body heat and brings down the body temperature to normal.
- iv. Therefore, we sweat during summer and heavy exercise.

22. What are symptoms of diseases like night blindness, rickets, beriberi, neuritis, pellagra, anaemia, scurvy?

- i. Night Blindness: Difficulty seeing in low-light conditions or at night.
- ii. Rickets: Softening and weakening of bones, leading to deformities, particularly in the legs, pelvis, and ribcage.
- iii. Beriberi: Muscle weakness and wasting.
- iv. Neuritis: Numbness, tingling, or burning sensations in the affected nerves.
- v. Pellagra: Dermatitis (skin rash) that appears in areas exposed to sunlight, such as hands, feet, neck, and face.
- vi. Anemia: Fatigue and weakness.
- vii. Scurvy: Swollen and bleeding gums.

23. What do you mean by coenzymes?

- i. Coenzymes are helpers that work with enzymes to make chemical reactions happen in our bodies.
- ii. They transfer specific chemical parts between molecules during metabolic processes, like NAD⁺ and FAD, which help with energy production and redox reactions.

24. Find the full forms of FAD, FMN, NAD, NADP

- i. FAD: Flavin Adenine Dinucleotide
- ii. FMN: Flavin Mononucleotide
- iii. NAD: Nicotinamide Adenine Dinucleotide
- iv. NADP: Nicotinamide Adenine Dinucleotide Phosphate

25. What happens to the cells of injured tissue?

The cells of the injured tissue start dividing and increase in numbers to repair and heal the injured tissue.

26. Whether new cells are formed during healing of wound?

Yes, the injured cells are restored by the formation of new cells

27. Do the plants get injured when do we pluck the flowers? How are those wounds healed?

- i. Yes, plants get injured when we pluck flowers.
- ii. The wounds are healed by formation of new cells by cell division.

28. How does the growth of any living organism occur? Does the number of cells in their body increase? If yes, how?

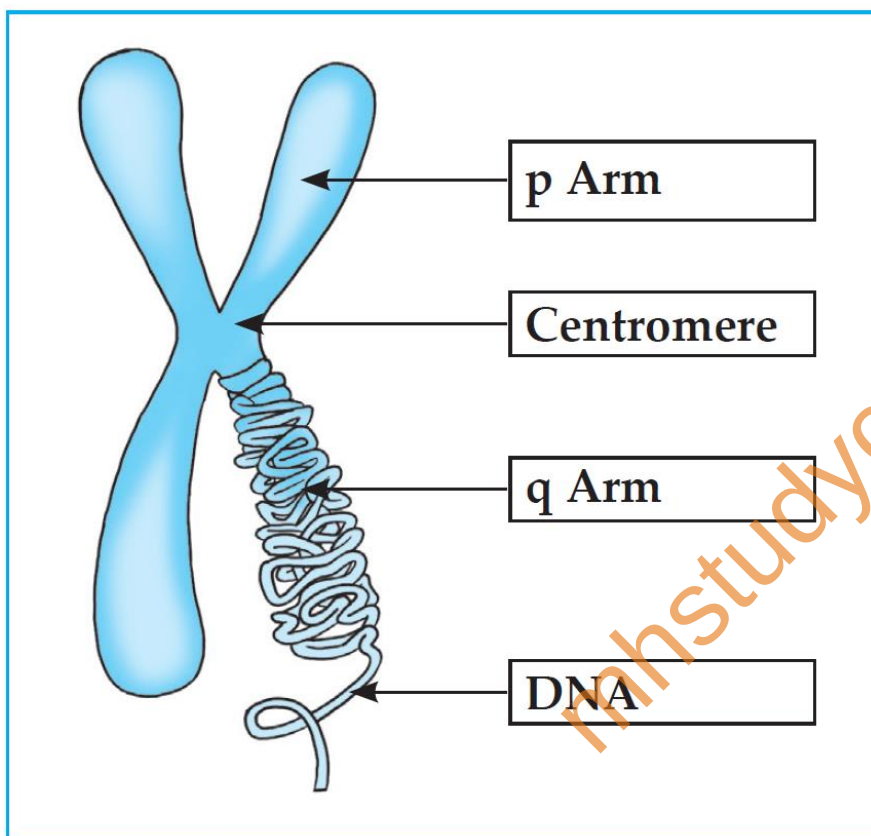
- i. Growth of any living organism occurs due to cell division.
- ii. Yes, the number of cells in their body increase.
- iii. The number of cells increases by the process of mitosis.

29. How is the new individual of a species formed from existing one of same species?

- i. New individual of a species is formed from the existing one of the same species by the process of reproduction.
- ii. It involves cell division - mitosis and meiosis.

30. What is the shape of chromosome? Give its names in the figure.

The shape of the chromosome is Sub-metacentric. (as one arm is slightly shorter than the other)



31. What do you mean by diploid ($2n$) cell?

A diploid cell is a cell that contains two sets of chromosomes. ($2n$)

32. What do you mean by haploid (n) cell?

Haploid cell is a cell that contains a single set of chromosomes. (n)

33. What do you mean by homologous chromosomes?

A pair of chromosomes that are similar by shape and organisation are called homologous chromosomes.

34. Whether the gametes are diploid or haploid? Why?

The gametes are haploid as they have a single set of chromosomes (n).

35. How are the haploid cells formed?

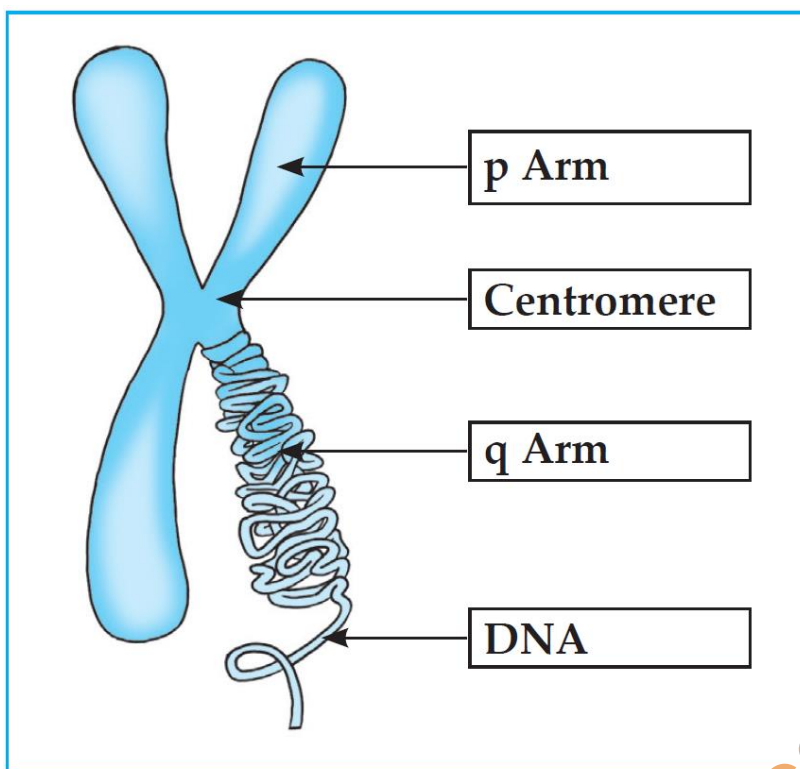
Haploid cells are formed by a type of cell division called meiosis.

36. What is the importance of haploid cells?

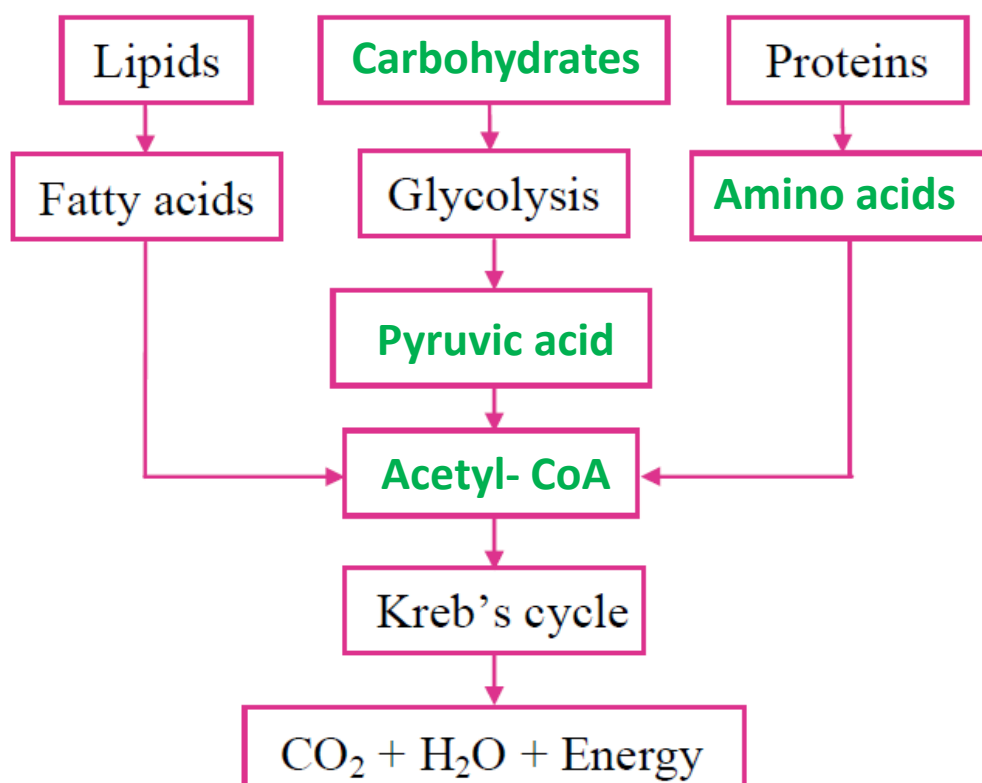
When haploid cells or gametes fuse, they form a diploid zygote so that the chromosome number in the organism is restored.

Previous Year Questions

1. Draw a neat diagram of the structure of chromosome and label the parts: a. Centromere b. p-arm.



2. Complete the following chart and state which process of energy production it represents:



Process of energy production through aerobic respiration of carbohydrates, proteins and fats is shown in the chart.

3. Match the columns:

	Column 'A'		Column 'B'
a.	Haemoglobin	1.	Skin
b.	Ossein	2.	Blood
		3.	Bones

a – 2, b - 3

4. a. Write main types of vitamins.

b. Name water soluble vitamins.

c. Name fat soluble vitamins.

- a. There are main six types of vitamins, e.g. A, B, C, D, E and K.
- b. B and C are water-soluble.
- c. A, D, E and K are fat-soluble

5. We feel exhausted after exercising.

- i. While exercising, the oxygen supply may not be enough for the need of muscles.
- ii. Hence, muscle cells perform anaerobic respiration.
- iii. As a result, less amount of energy is produced in our body and lactic acid accumulates due to which we feel tired.

6. Distinguish between Aerobic respiration and Anaerobic respiration (any two points).

Exercise Question 3 – C

7. Cell division is one of the very important properties of cells and living organisms.

Exercise Question 4 – C

8. Differentiate between mitosis and meiosis. (Any 3 points)

Exercise Question 3 – B

9. What are vitamins? Write two types of vitamins.

- i. Vitamins are a group of heterogeneous compounds of which, each is essential for proper operation of various processes in the body.
- ii. There are main six types of vitamins, e.g. A, B, C, D, E and K.
- iii. Out of these, A, D, E and K are fat-soluble whereas B and C are water-soluble.

10. Sometimes higher plants and animals too perform anaerobic respiration.